RITVIK AGGARWAL

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EDUCATION & HONORS

Indiana University, Luddy School of Informatics, Computing and Engineering *May 2027*

Bachelor of Science in Informatics, Minor in Business and Entrepreneurship — GPA: 3.85/4.00

- Dean's List; member of Alpha Lambda Delta and the National Society of Collegiate Scholars for academic standing
- Selected for the Luddy Academic Integrity Council; two-time competitor at the National Swimming Championship

TECHNICAL SKILLS

Python, SQL, Java, LangChain, Mistral AI, Microsoft Graph API, RAG, TensorFlow, PyTorch, AWS, Azure OpenAI, Docker, Git

PROFESSIONAL EXPERIENCE

EXL | Analytics and AI Solutions Intern *June 2026 – Present*

New York City, NY

- Building an Outlook add-in with Microsoft Entra ID authentication, letting Microsoft Graph securely verify each user before pulling their email data
- Retrieving and filtering a user's Outlook emails by day, week, or month through Microsoft Graph, scanning an average of 100 emails per session
- Connecting LangChain to Mistral AI to convert filtered emails into prioritized, due-dated action items, extracting an average of 50 tasks per week
- Building the Express backend that turns Mistral AI's structured output into a live task tracker inside Outlook, so users never leave their inbox

Handshake | AI Trainee *December 2025 – Present*

Bloomington, IN

- Evaluated AI-generated responses to real conversation prompts, judging tone, personality, and presentation for context
- Assessed personality and formatting as separate dimensions, flagging responses that were warm but hard to scan or polished but impersonal
- Documented judgments using exact quotes and specific examples to explain what worked, what did not, and why it mattered
- Contributed structured evaluations across multiple task types to help train AI systems to follow formatting instructions more accurately

School of AI | Machine Learning Intern *December 2025 – April 2026*

Bloomington, IN

- Building an end-to-end churn prediction pipeline using tenure, billing, and contract features, improving model F1 score through iterative tuning
- Automating F1 and precision-recall evaluation across 3 ML pipelines, catching model degradation before it reached production
- Handling severe class imbalance in a fraud detection model with class weighting and undersampling, improving detection of rare fraud cases
- Delivering 25 ML projects during the internship, reusing a consistent preprocessing and evaluation pipeline to speed up every new build

Deloitte | Gen AI Intern *May 2025 – August 2025*

Gurugram, India

- Building the backend pipeline that parses uploaded PDF, Word, and Excel files, extracting text for the AI model to answer user questions
- Chunking and embedding extracted document text into a vector database, enabling fast retrieval across a growing document collection
- Building retrieval logic that matches user questions to the closest document chunks, grounding AI answers instead of relying on guesses
- Improving backend logic and frontend responsiveness, boosting overall application performance by 30% for enterprise clients

PROJECTS

AI Trading & Behavior Modeling *Spring 2026*

- Built an LSTM-Transformer ensemble in PyTorch, fusing bidirectional LSTM and positional-encoded Transformer outputs to predict returns
- Backtested signals out-of-sample with configurable transaction costs, tracking Sharpe ratio, drawdown, hit rate, and a confusion matrix
- Built a bidirectional LSTM with attention pooling in PyTorch, classifying user state into 4 categories from 10 behavioral features
- Added an HMM-style transition layer to smooth predictions across a session, then deployed a live FastAPI demo on Render

Discord Server Memory Bot *Spring 2025*

- Built a Discord bot in Node.js that classifies incoming messages with a Naive Bayes model and saves important ones to SQLite
- Built a custom retrieval ranking function scoring token overlap, keyword coverage, and recency to surface relevant saved messages
- Connected retrieved memories to Mistral AI to generate cited, source-linked answers to natural language questions about server history
- Added a regex-based privacy filter blocking passwords, API keys, and card numbers, plus user-level commands to view and delete stored data

LEADERSHIP & INVOLVEMENT

Professional Readiness at IU | Co-President *August 2024 – December 2024*

Bloomington, IN

- Led 10+ workshops on business etiquette and professional development for 30+ students each semester, covering interviews and networking
- Recruited guest speakers from 5+ industries for sessions with 50+ participants, and managed fundraising for venues and materials

Middle Eastern Business Association | Co-President *January 2024 – Present*

Bloomington, IN

- Partnered with 5+ campus organizations to co-host inclusive cultural and business events, coordinating logistics and cross-club scheduling
- Led marketing and outreach efforts that grew club membership and attendance through social media campaigns and in-person tabling

INTERESTS

Computer Vision | Swimming | Badminton | Traveling | Vibe Coding | Automation